

DIGT 3107 SECTION A

Phase 1 Submission

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Understanding the Domain

Elmwood University operates a single guest house used by visiting lecturers, parents, alumni, and event guests. The current booking process relies on manual records and email communication, which has resulted in double bookings, lost requests, and poor coordination. The proposed Room/Hotel Booking System will provide a digital database for managing guests, rooms, bookings, cancellations, payment methods, and check-in/check-out information.

Functional Requirements

- List available rooms for a given date range.
- Show all bookings for a specific guest.
- Identify the most-booked room types.
- Calculate the total number of bookings and revenue for 2025, including the required discount.
- Calculate revenue per room type for a given month, including only completed bookings.
- Track cancellation trends by month or room type.

Entities

- **Guest:** stores the person making/staying under a booking.
- **Room:** Room: stores each actual physical room, including room number and a link to its room type.
- **RoomType:** stores the categories Single, Duplex, and Presidential Suite, along with their capacity and nightly price.
- **Booking:** connects a guest to a room. It stores the booking date/time, check-in and check-out date/time, payment method, and a status value (Active, Completed, or Cancelled) that tracks whether the stay is still upcoming, finished, or was cancelled.

Attributes and Keys

Entity	Attributes
Guest	GuestID PK , FirstName, LastName, Email, Phone, IsUniversityStaff
RoomType	RoomTypeID PK , TypeName, Capacity, NightlyPrice
Room	RoomID PK , RoomNumber, RoomTypeID FK
Booking	BookingID PK , GuestID FK , RoomID FK , BookingDateTime, CheckInDateTime, CheckOutDateTime, PaymentMethod, Status (allowed values: Active, Completed, Cancelled), CancellationDateTime

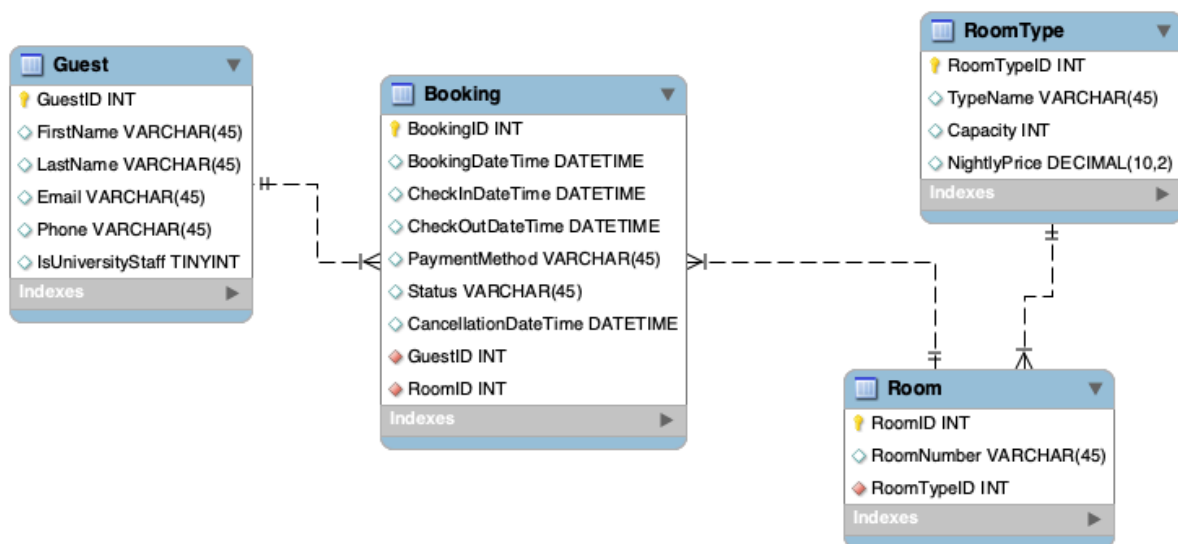
Relationships

The database uses three one-to-many relationships. One Guest may have many Bookings, while each Booking belongs to one Guest. One RoomType may describe many Rooms, while each Room belongs to one RoomType. One Room may appear in many Bookings over time, while each Booking is assigned to one Room. One RoomType may describe many Rooms, while each Room belongs to one RoomType.

- Guest → Booking = one-to-many
 - Guest has many Bookings
- RoomType → Room = one-to-many
 - RoomType has many Rooms
- Room → Booking = one-to-many
 - Room has many Bookings

ER Diagram

The following ER diagram shows the logical database design, including entities, primary keys, foreign keys, and relationships.



Assumptions, Clarifications, Business Rules

The database design assumes that each booking is associated with one guest and one room, while a guest may have multiple bookings over time. Each room belongs to one room type, and each room type may be associated with multiple rooms.

Payment information is stored as a payment method within the Booking entity because the requirements only specify credit card, debit card, or cash and do not describe separate payment transactions, refunds, or multiple payments.

Cancellation information is stored within the Booking entity using a booking status and cancellation date/time rather than as a separate entity. This allows cancellation trends to be tracked while keeping the model simple.

The Status field is limited to three values: Active, Completed, and Cancelled. This distinction is needed because the monthly revenue requirement only counts completed bookings, and to tell a finished stay apart from one that is still in progress. CancellationDateTime is only filled in when Status is Cancelled, so the two fields stay consistent with each other.

The requirements state that Elmwood University staff receive a 10% discount, so the Guest entity includes an IsUniversityStaff attribute. However, the requirement to calculate 2025 revenue says a 10% discount should be applied to all room prices. We interpreted the staff discount as applying only to university staff and note this as an ambiguity that would require clarification in a real requirements meeting.

The system does not allow overbooking. A room cannot have overlapping active bookings for the same date range, although this business rule is not directly enforced by the ER diagram itself.